

**ESCOLA
SUPERIOR
DE TECNOLOGIA
E GESTÃO**

P.PORTO

Data Warehousing

Data Warehouse

- A subject-oriented, integrated, time-variant, and non-volatile collection of data in support of management's decision-making process (Inmon, 1993).

Comparação de Sistemas Transacionais e Data Warehousing

TABLE 31.1 Comparison of OLTP systems and data warehousing systems.

CHARACTERISTIC	OLTP SYSTEMS	DATA WAREHOUSING SYSTEMS
Main purpose	Support operational processing	Support analytical processing
Data age	Current	Historic (but trend is toward also including current data)
Data latency	Real-time	Depends on length of cycle for data supplements to warehouse (but trend is toward real-time supplements)
Data granularity	Detailed data	Detailed data, lightly and highly summarized data
Data processing	Predictable pattern of data insertions, deletions, updates, and queries. High level of transaction throughput.	Less predictable pattern of data queries; medium to low level of transaction throughput
Reporting	Predictable, one-dimensional, relatively static fixed reporting	Unpredictable, multidimensional, dynamic reporting
Users	Serves large number of operational users	Serves lower number of managerial users (but trend is also toward supporting analytical requirements of operational users)

Problemas

TABLE 31.2 Problems of data warehousing.

Underestimation of resources for data ETL

Hidden problems with source systems

Required data not captured

Increased end-user demands

Data homogenization

High demand for resources

Data ownership

High maintenance

Long-duration projects

Complexity of integration

Arquitetura de um SDW

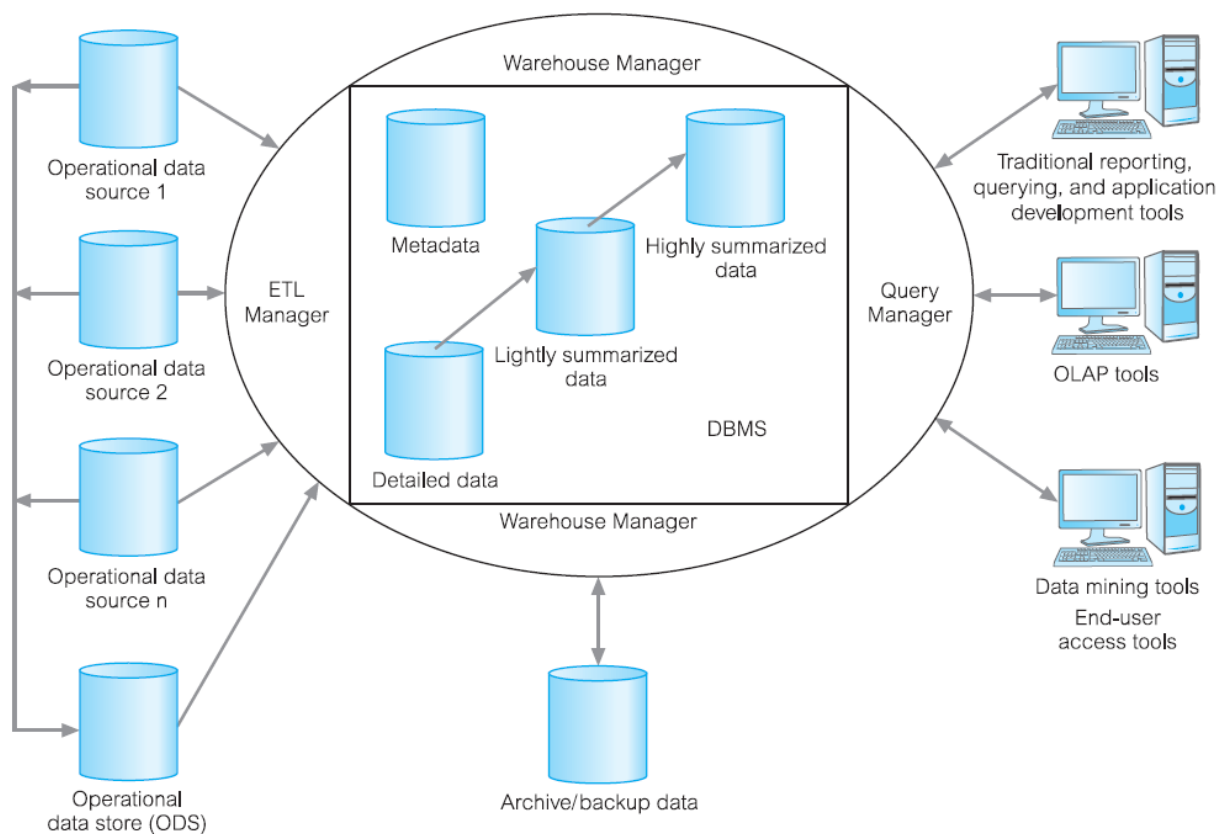


Figure 31.1 Typical architecture of a data warehouse.

Metodologias

TABLE 32.1 The main advantage and disadvantage associated with the development of an EDW using Inmon's CIF methodology and Kimball's Business Dimensional Lifecycle.

METHODOLOGY	MAIN ADVANTAGE	MAIN DISADVANTAGE
Inmon's Corporate Information Factory	Potential to provide a consistent and comprehensive view of the enterprise data.	Large complex project that may fail to deliver value within an allotted time period or budget.
Kimball's Business Dimensional Lifecycle	Scaled-down project means that the ability to demonstrate value is more achievable within an allotted time period or budget.	As data marts can potentially be developed in sequence by different development teams using different systems; the ultimate goal of providing a consistent and comprehensive view of corporate data may never be easily achieved.

Kimball's Business Dimensional Lifecycle

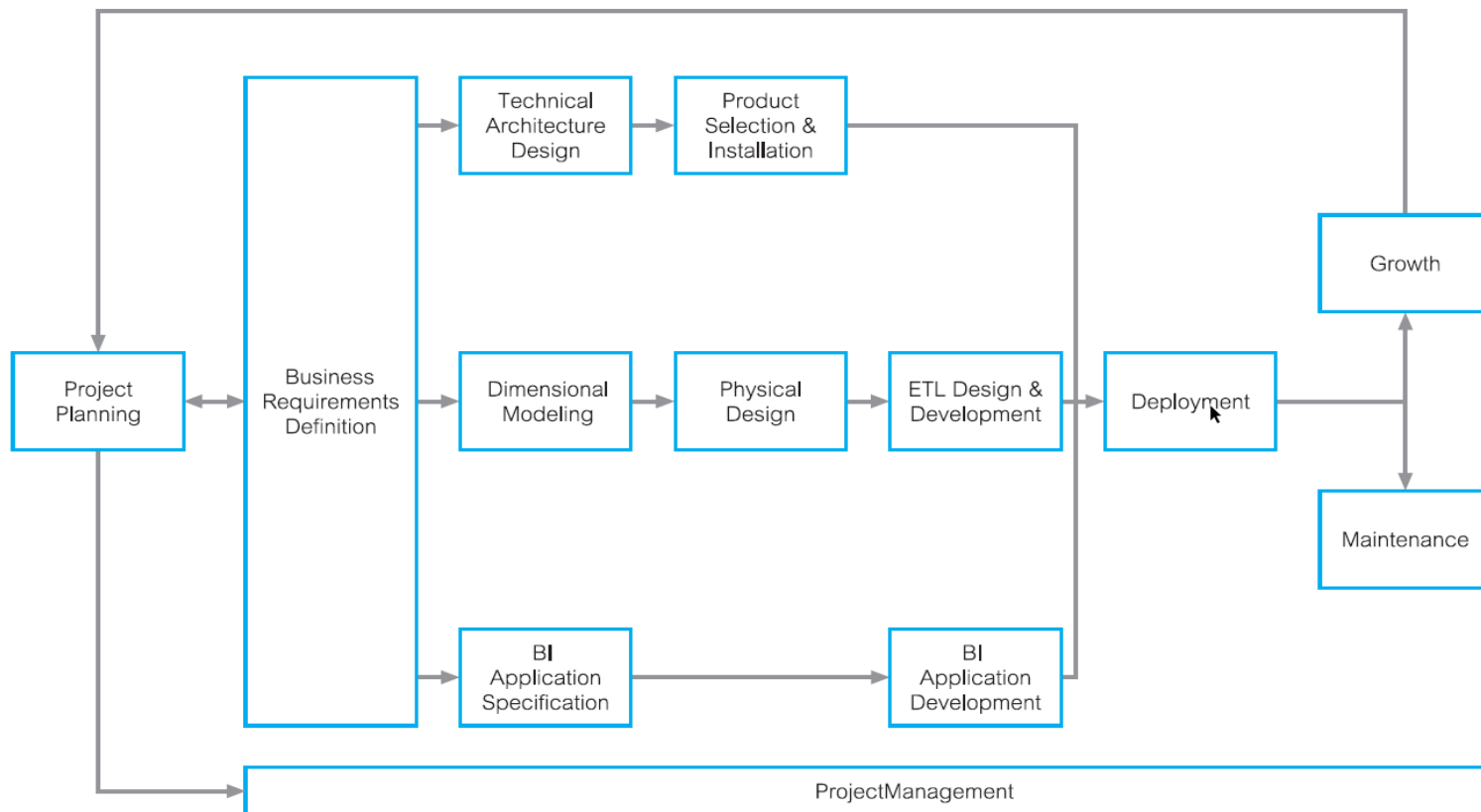


Figure 32.1 The stages of Kimball's Business Dimensional Lifecycle (Kimball, 2008).

Modelo Dimensional

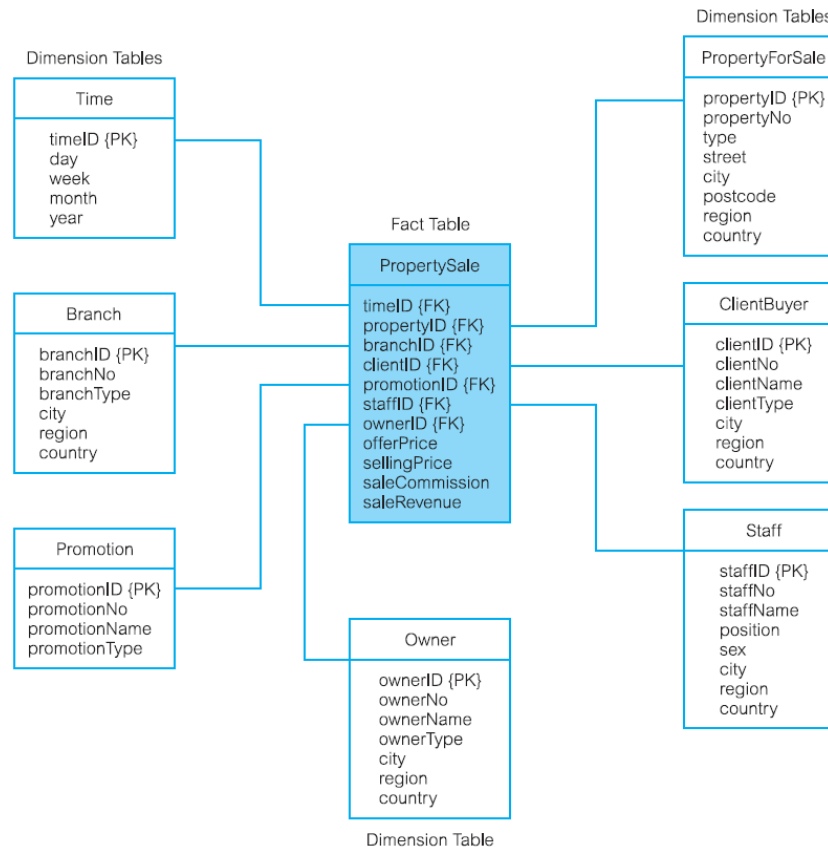


Figure 32.2 Star schema (dimensional model) for property sales of *DreamHome*.

Modelo ER vs Dimensional

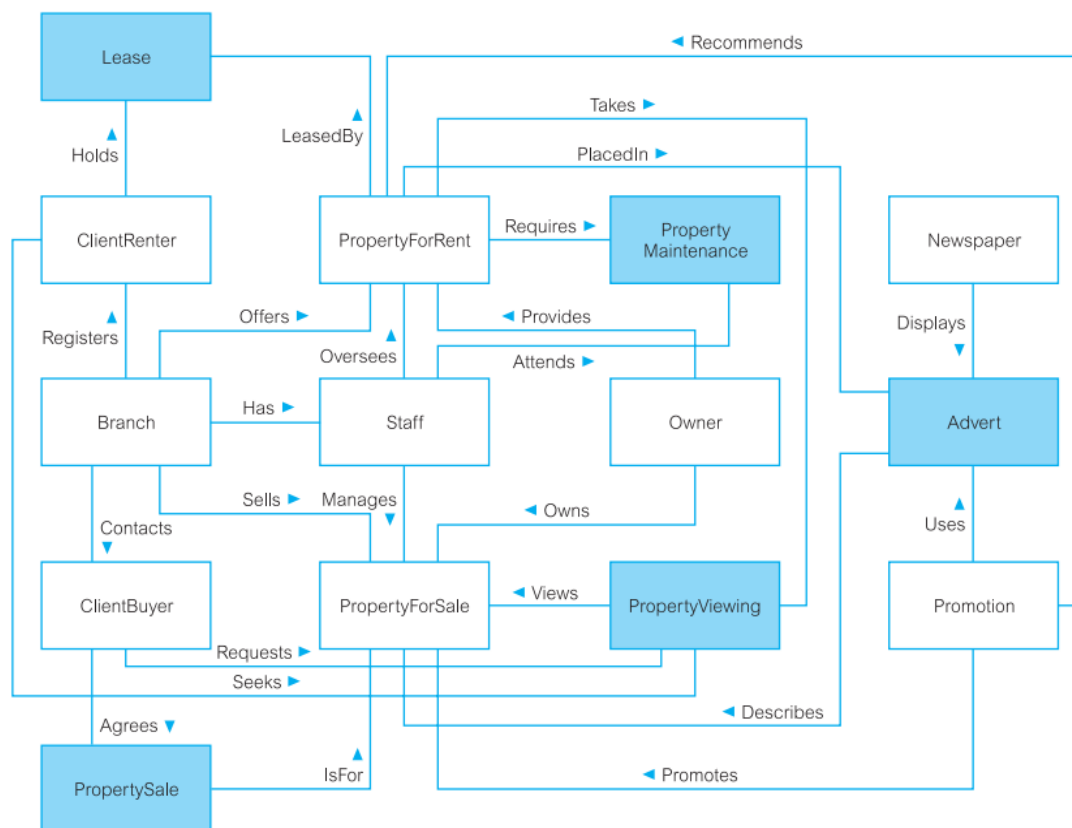


Figure 32.5 ER model of an extended version of *DreamHome*.

Modelo ER vs Dimensional

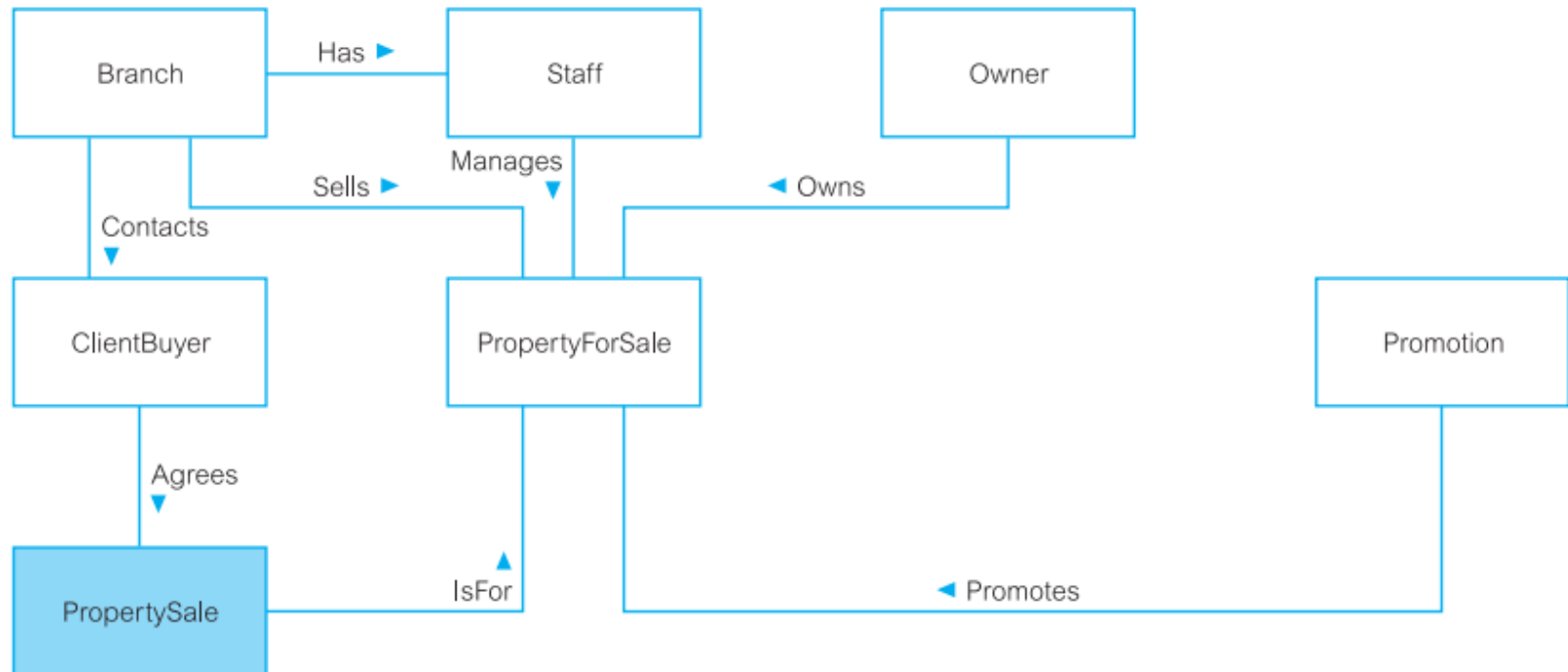


Figure 32.6 Part of ER model in Figure 32.5 that represents the data requirements of the property sales business process of *DreamHome*.

Modelo ER vs Dimensional

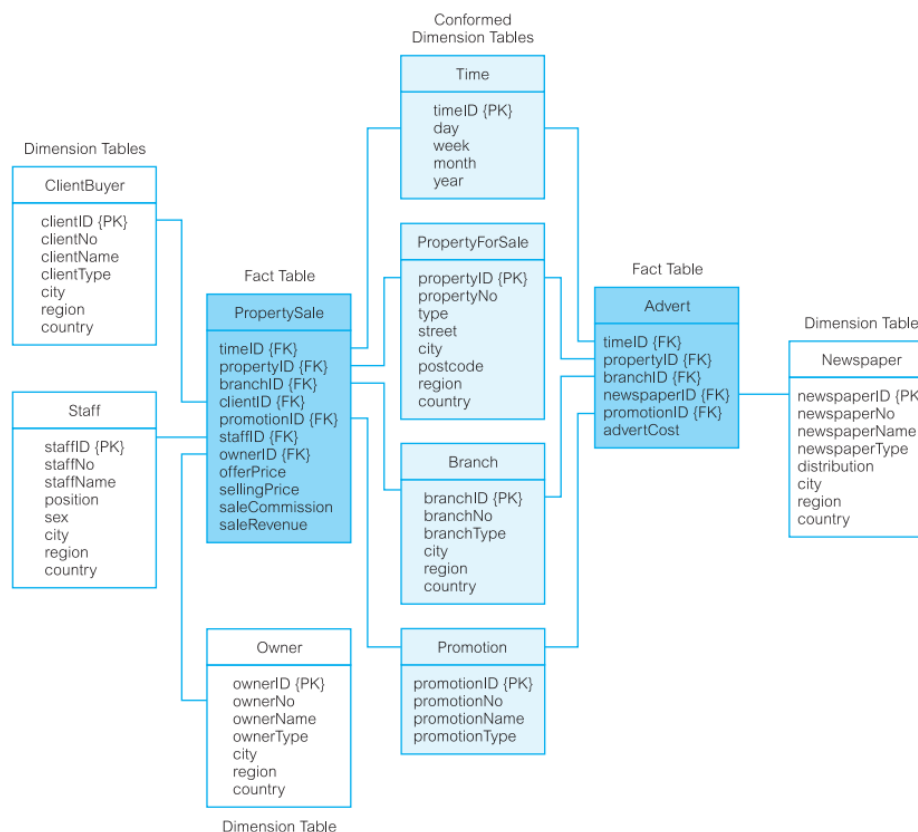


Figure 32.7 Dimensional model for property sales and property advertising with Time, PropertyForSale, Branch, and Promotion as conformed (shared) dimension tables.

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